Cache performance

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Computer Organization and Design, 4th Edition

1. Marks
2. Homework
3. Test 2 - 27.05, 13.30 - 14.00
4. Test 2 - 27.05, 14.30 - 15.00
5. S14 - **retake**, 17 - 19
6. Cache performance

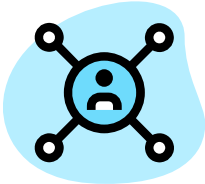
Important formulas

n

Memory size = 2^n blocks

m

Cache size = 2^m blocks
m bits for index



Tag size

No. of bits/word = 32
32 - n - block offset - word offset

1 B = 8 b
(1 byte = 8 biți)

Important formulas

TAG	INDEX	BLOCK OFFSET	WORD OFFSET
32-m-b.o.-w.o	m	$\log_2$(no. of **words / block**)	$\log_2$(no.of **bytes / word**)

n

Memory size = 2^n blocks

m

Cache size = 2^m blocks
m bits for index

Tag size

No. of bits/word = 32
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Word addressed vs. byte addressed

Important formulas

CPU clk cycles

CPU exec time

CPU clk cycles = CPU exec. clk cycles + Mem. stall cycles

Important formulas

CPU clk cycles

CPU clk cycles = CPU exec. clk cycles + Mem. stall cycles

CPU exec time

CPU time = (CPU exec. clk cycles + Mem. stall cycles) x Clk cycle time

Important formulas

Mem. stall cycles

Mem. stall cycles = Read stall cycles **+** Write stall cycles

Read stall cycles

Read stall cycles = **Reads / program** * read miss rate x read miss penalty

Write stall cycles

Write stall cycles = **Writes / program** * write miss rate x write miss penalty

Mem. stall cycles

Mem. stall cycles = **Mem. accesses / program** * Miss rate x Miss penalty

Computing miss cycles

Instr. miss cycles

Instr. miss cycles = **IC** x instr. miss rate x miss penalty

Data miss cycles

Data miss cycles = **IC** x **data access rate** x data miss rate x miss penalty

Mem. stall cycles = Instr. miss cycles + Data miss cycles

Problem 1

Assume we have computer where the cycles per instruction is *1.0* when all memory accesses hit in the cache.
The only data accesses are loads and stores, and these total *50%* of the instructions. If the miss penalty is 25 clock cycles and the miss rate is 2%, how much faster would the computer be if all instructions were cache hits?

- Compute: CPU time ideal cache, Mem. stall cycles, CPU time real cache
- Divide the performance of the 2 scenarios

Important formulas

AMAT = Average Memory Access Time

AMAT = **%instr** x (**Hit time** + Instr. Miss Rate x Miss Penalty) +
%data x (**Hit time** + Data Miss Rate x Miss Penalty)

AMAT = **Hit time** + Miss Rate x Miss Penalty

Problem 2

Assume the miss rate of an instruction cache is 2% and the miss rate of the data cache is 4%. If a processor has a CPI of 2 without any memory stalls and the miss penalty is 100 cycles for all misses, determine how much faster a processor would run with a perfect cache that never missed. Assume the frequency of all loads and stores is 36%.

- Compute: Instr. miss cycles, Data miss cycles, CPI (including memory stall)
- Divide the performance of the 2 scenarios

Thanks!